SIMATS ENGINEERING

ASSESSMENT TOOL 2 (CO2 - AT2)

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Register Number	1921260256
Course Code	CSA0709
Course Name	Computer Networks
Assessment	Scenario-Based Activities (5 questions x 6 marks = 30 marks)
Weightage	30%

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Code of Conduct: I, Bhavatharani H (1921260256), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Question 1 (6 marks)

A newly established Smart Campus is deploying a secure network infrastructure. The administrator has the block **192.168.100.0/24**. Assigned masks: School of Computing /25, School of Electronics /26, School of Mechanical Engineering /27. Determine (1) network and broadcast address of each department, (2) valid host range, (3) usable hosts per subnet, (4) remaining unused range.

Answer

Approach: Masks: /25 = 255.255.255.128, /26 = 255.255.255.192, /27 = 255.255.255.224 (block sizes 128, 64, 32 = 256 - last mask octet). Subnets are allocated one after another, largest first, starting at 192.168.100.0 so they do not overlap. Usable hosts = 2^h - 2, where h = host bits.

Department	Mask	Network address	Broadcast address	Valid host range	Usable hosts
School of Computing	/25 (.128)	192.168.100.0	192.168.100.127	192.168.100.1 - 192.168.100.126	2^7 - 2 = 126
School of Electronics	/26 (.192)	192.168.100.128	192.168.100.191	192.168.100.129 - 192.168.100.190	2^6 - 2 = 62
School of Mechanical Engg.	/27 (.224)	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222	2^5 - 2 = 30

Unused range (Task 4): The three subnets use 128 + 64 + 32 = 224 addresses (192.168.100.0 - 192.168.100.223). The remaining **192.168.100.224 - 192.168.100.255** (192.168.100.224/27, 32 addresses; hosts .225 - .254 usable) is free for future expansion.

Conclusion: The allocation is contiguous, non-overlapping and fits inside the /24 with 32 addresses spare.

Question 2 (6 marks)

A software company has the private block **10.10.0.0/16**. Masks: Software Development /24, Quality Assurance /25, Human Resources /26, Executive Management /27. Determine (1) network and broadcast addresses, (2) valid host ranges, (3) usable hosts, (4) unused range remaining in the block.

Answer

Approach: Allocate sequentially from 10.10.0.0, largest subnet first (this keeps the addressing contiguous and avoids wasted gaps).

Department	Mask	Network address	Broadcast address	Valid host range	Usable hosts
Software Development	/24 255.255.255.0	10.10.0.0	10.10.0.255	10.10.0.1 - 10.10.0.254	2^8 - 2 = 254
Quality Assurance	/25 255.255.255.128	10.10.1.0	10.10.1.127	10.10.1.1 - 10.10.1.126	2^7 - 2 = 126
Human Resources	/26 255.255.255.192	10.10.1.128	10.10.1.191	10.10.1.129 - 10.10.1.190	2^6 - 2 = 62
Executive Management	/27 255.255.255.224	10.10.1.192	10.10.1.223	10.10.1.193 - 10.10.1.222	2^5 - 2 = 30

Unused range (Task 4): Addresses used = 256 + 128 + 64 + 32 = 480 (10.10.0.0 - 10.10.1.223). The remaining range is **10.10.1.224 - 10.10.255.255**, i.e. 65,536 - 480 = **65,056 addresses** (starts with 10.10.1.224/27, then 10.10.2.0 onwards) available for future departments.

Conclusion: Only about 0.73% of the /16 is used, leaving ample room for growth.

Question 3 (6 marks)

A smart hospital is migrating from IPv4 to IPv6. Assigned network: **2001:0db8:abcd:0000:0000:0000:0000:0000/64**. (1) Compress the address, (2) expand it back to full form, (3) identify the network prefix, (4) calculate the number of host addresses in the subnet, (5) state the total possible host addresses in this network.

Answer

Task	Result
1. Compressed notation	Drop leading zeros in each group: 2001:db8:abcd:0:0:0:0:0. Replace the longest run of all-zero groups with '::' (once only): **2001:db8:abcd::/64**
2. Expanded (full) form	Restore the omitted zero groups so there are 8 groups of 4 hex digits: **2001:0db8:abcd:0000:0000:0000:0000:0000**
3. Network prefix (/64)	The first 64 bits (first 4 groups) are the network part: **2001:0db8:abcd:0000** = 2001:db8:abcd:0::/64. The last 64 bits (last 4 groups) are the interface identifier (host part).
4. Host addresses in the subnet	Host bits = 128 - 64 = 64, so hosts = 2^{64} = **18,446,744,073,709,551,616** (about 1.84 x 10^{19}).
5. Total possible addresses in this network	Same calculation: $2^{(128-64)}$ = **2^{64} = 18,446,744,073,709,551,616** addresses, from 2001:db8:abcd::0 to 2001:db8:abcd:0:ffff:ffff:ffff:ffff.

Note: IPv6 has no broadcast address, so the '-2' rule of IPv4 does not apply. The all-zeros interface ID (2001:db8:abcd::) is reserved as the Subnet-Router anycast address, so strictly 2^{64} - 1 addresses are assignable to hosts.

Conclusion: A single /64 easily supports thousands of medical devices and IoT sensors, with room for sub-division by assigning multiple /64 subnets (for example per ward) from the hospital's larger allocation.

Question 4 (6 marks)

An ISP core router has routing table entries **192.168.1.0/24, 192.168.2.0/24, 192.168.3.0/24**. A packet arrives with destination **192.168.2.45**. Determine (1) the destination subnet, (2) the matching routing entry, (3) the next-hop network/interface, (4) the default route used if nothing matches.

Answer

Step 1 - Compare the destination with each entry (AND with the /24 mask 255.255.255.0):

Routing entry	Address range	Does 192.168.2.45 fall in it?
192.168.1.0/24	192.168.1.0 - 192.168.1.255	No (third octet 2 is not 1)
192.168.2.0/24	192.168.2.0 - 192.168.2.255	**Yes** (192.168.2.45 AND 255.255.255.0 = 192.168.2.0)
192.168.3.0/24	192.168.3.0 - 192.168.3.255	No (third octet 2 is not 3)

Task	Result
1. Destination subnet	**192.168.2.0/24** (network 192.168.2.0, broadcast 192.168.2.255; 192.168.2.45 is a valid host).
2. Matching routing entry	**192.168.2.0/24**. It is the only match, so it is also the longest-prefix match.
3. Next hop / interface	The packet is forwarded out of the router interface connected to the 192.168.2.0/24 customer network (e.g. Gi0/2 - interface name assumed, as the table gives none). Because the network is directly connected, the router ARPs for 192.168.2.45 and delivers the frame directly; if it were learned from another router, that router's address would be the next hop.
4. Default route	**0.0.0.0/0** (mask 0.0.0.0) pointing to the ISP upstream router/interface, e.g. `ip route 0.0.0.0 0.0.0.0 <upstream next-hop>`. It is used only when no more specific entry matches (for example 192.168.5.10); without it such packets are dropped.

Conclusion: The router uses longest-prefix match, finds 192.168.2.0/24, and forwards the packet out of the corresponding interface.

Question 5 (6 marks)

Two departments are connected through a Layer-3 router: Administration LAN **192.168.10.0/26** (gateway 192.168.10.1) and Finance LAN **192.168.10.64/26** (gateway 192.168.10.65). Determine (1) network and broadcast addresses, (2) valid host ranges, (3) suitable IPs for three computers in each LAN, (4) the default gateway for hosts in each LAN.

Answer

Approach: /26 = 255.255.255.192, block size 64, so each LAN has 2^6 - 2 = 62 usable hosts.

Item	Administration LAN	Finance LAN
Network address	192.168.10.0	192.168.10.64
Broadcast address	192.168.10.63	192.168.10.127
Valid host range	192.168.10.1 - 192.168.10.62	192.168.10.65 - 192.168.10.126
Usable hosts	62	62
Subnet mask	255.255.255.192	255.255.255.192
Router interface / default gateway	192.168.10.1	192.168.10.65
PC1	192.168.10.2	192.168.10.66
PC2	192.168.10.3	192.168.10.67
PC3	192.168.10.4	192.168.10.68

Decisions and rationale:

- Host addresses are taken from within each LAN's valid range, skipping the gateway address (.1 / .65), so there is no IP conflict.
- Hosts in Administration use **192.168.10.1** and hosts in Finance use **192.168.10.65** as default gateway, because each gateway lies in the same subnet as its hosts and is the router interface for that LAN.
- When Administration PC1 (192.168.10.2) sends to Finance PC1 (192.168.10.66), the destination is outside its /26, so the packet goes to gateway 192.168.10.1; the router routes it to the Finance interface 192.168.10.65, which delivers it to the host.

Conclusion: The two /26 subnets are contiguous, non-overlapping and correctly separated by the Layer-3 router.

Marks: 5 questions x 6 marks = 30 marks.